

Paola Romero

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EDUCATION

Massachusetts Institute of Technology (MIT)

Class of 2027

Cambridge, MA

- Candidate for B.S. in Mechanical Engineering. GPA: 4.7/5.0
- Involvements: Pappalardo Lab Apprentice, Edgerton Center Student Project Lab Asst. Shop Manager, Electronics for Mechanical Engineers Lab Assistant, Climbing Club, Wind Ensemble, Spinning Arts club
- Relevant Courses: Product Design, Design and Manufacturing I & II, Dynamics and Controls I & II, Thermal-Fluids Engineering I & II, Mechanics and Materials I & II, Numerical Computation, Differential Equations

EXPERIENCE

Lexington Medical

May. 2026-Aug. 2026

Cambridge, MA

Robotic Surgical Stapling - R&D Mechanical Engineer

- Characterized mechanical properties of strings/wires through ~300 loading/unloading cycles and 37 break tests
- Developed a test fixture for a 2-DOF mechanism; results informed motor requirements and articulation performance.
- Designed a 4-channel sensor testing PCB with Wheatstone bridges, amplification, temperature compensation, filtering, and power regulation.

Fabrication Integrated Design Lab (FIDL)

Sep. 2025-May. 2026

Cambridge, MA

Soft Robotics - Undergraduate Researcher

- Designed a bidirectional, tendon-driven grasper with both sequential and linked-joint actuation, enabling complex grasping tasks.
- Developed a model using pseudo-rigid body modeling to characterize the grasper's bending.
- Awarded 3rd Place for Undergraduate Science at 2026 MIT MechE De Florez Competition, presented at 2026 New England Manipulation Symposium, and currently preparing a conference paper submission.

Center of Innovation and Manufacturing at UPC

Jun. 2025-Aug. 2025

Barcelona, Spain

Additive Manufacturing - Research Intern

- Designed and prototyped a 6 m scale model of a 24 m post-tensioned concrete bridge in SolidWorks, including FEA simulations and hand calcs to validate performance under a 200 kN tensioning force and live loads.
- Created a novel modular mating system to join four 1.5m bridge sections

Campos Engineering

Jun. 2024-Aug. 2024

Dallas, TX

HVAC - Innovation Intern

- Assembled and tested dozens of SMT PCBs, cutting debugging time by 52% and improving quality control.
- Designed "Tulip," an electronics enclosure for an air velocity grid, improving airflow and reducing assembly time.
- Built a custom MIG-welded steel lift for an air velocity grid, rated for 40 kg load, ensuring durability and safety.
- Created the gateway PCB in KiCad—schematic, layout, routing, and component selection for signal reliability.

Campos Engineering

Jun. 2023-Jul. 2023

Dallas, TX

Innovation Intern

- Programmed Digi XBee 3 modules in MicroPython for temperature/humidity monitoring; soldered 30+ sensor PCBs and identified HVAC malfunctions in Campos Engineering's Dallas office.
- Iterated on 1mm wide airtight seals for airflow instruments, reducing manufacturing cost by 95%.
- Designed a 3D-printed wireless pressure sensor case, cutting sensor cost by 68% and streamlining assembly.
- Developed 3 unique prototypes for custom tools and components.

SKILLS

- **Technical:** Additive Manufacturing (SLA & FDM), Arduino & Raspberry Pi Microcontrollers, CNC Machining (Mill), Manual Lathe, MIG Welding, Injection Molding, Sheet forming (Metal and Thermoplastic), and Power Tools
- **Software:** CAD (SolidWorks, Fusion 360, Onshape), FEA (SolidWorks), CAM (Fusion360), MATLAB, Arduino (C++), KiCad, Python, HTML, CSS, Javascript